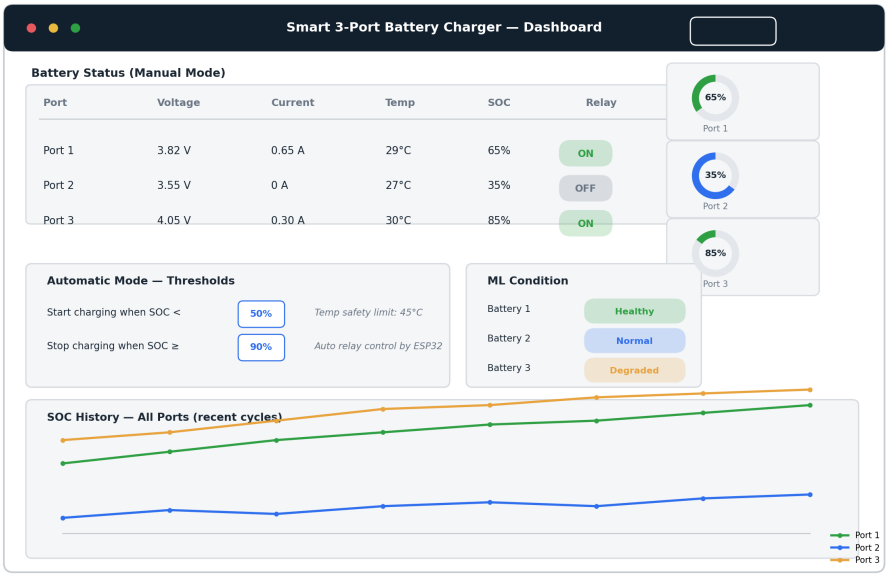
PROJECT PROPOSAL

Smart Multi-Port Battery Charging & Management System Using ESP32 and Machine Learning

A 3-port intelligent battery charger with CC/CV charging, ESP32-based monitoring, and ML-based condition assessment



Submitted by

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1. Full Concept

The proposed system is a smart multi-port battery charging, monitoring, and management system designed to charge and monitor several rechargeable batteries independently. The system uses an AC input power source, a suitable AC-to-DC power supply/charging stage, independent charging ports, relays, voltage/current/temperature sensors, and an ESP32 microcontroller.

Each battery port is monitored separately. The ESP32 continuously collects battery voltage, charging/discharging current, temperature, charging time, discharging time, charge delivered, discharge capacity, and cycle information. The measured data are sent to a laptop or smartphone through Bluetooth/Wi-Fi. A software dashboard displays the real-time condition of every battery and provides control over each charging port.

The system has two operating modes: Manual Mode and Automatic Mode. In Manual Mode, the user can turn each charging port ON or OFF from the software. In Automatic Mode, the user sets charging limits — for example, starting charging when SOC falls below 50% and stopping charging at 90% — and the ESP32 automatically controls the corresponding relay.

The system also incorporates Machine Learning (ML). Instead of using ML only for SOC estimation, the model uses several parameters — voltage, current, temperature, charging/discharging time, delivered capacity, cycle number, and charging/discharging characteristics — to evaluate battery condition, usable capacity, degradation, abnormal behaviour, and health trend. This makes the project more than a conventional battery charger.

> **Important:** *If the battery is a 3.7 V Li-ion cell, a fixed 3.7 V supply cannot be connected directly to it for charging. The charging stage must provide the appropriate CC/CV charging profile with approximately 4.2 V termination for a single Li-ion cell. The relay only controls whether that charging channel is connected or disconnected.*

2. System Architecture

The diagram below shows the complete signal and power flow of the system, from the AC mains input through to ML-based data analysis on the dashboard. AC power is protected, converted to regulated DC rails, and routed through a CC/CV charging stage; each of the three ports is switched by its own relay, sensed independently, and read by the ESP32, which also relays measurements to the dashboard over Bluetooth/Wi-Fi and receives Automatic-mode threshold commands back.

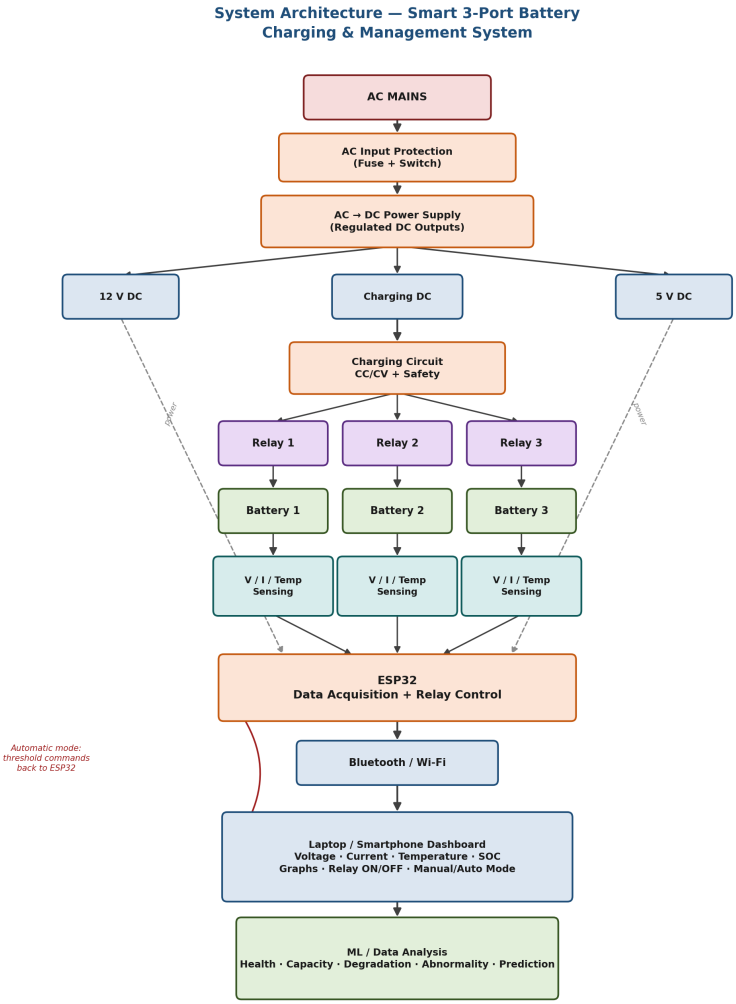


Figure 1: Full system architecture — AC mains to ML-based data analysis.

3. Working Principle / Step-by-Step Flow

Step 1 — AC Power Input

The system receives power from the AC mains through a suitable isolated AC-to-DC power supply system.

Step 2 — DC Power Conversion

The AC is converted into regulated DC outputs required by different parts of the system, for example:

- 12 V DC → system / charging-stage supply
- 5 V DC → ESP32 / peripherals or auxiliary circuits
- Appropriate regulated charging voltage/current → battery charging channels

The exact charging voltage depends on the selected battery chemistry.

Step 3 — Independent Charging Ports

Each battery has its own charging channel. For a 3-port system: Port 1 → Battery 1, Port 2 → Battery 2, Port 3

→ Battery 3. Each port contains its own relay/switch and sensing circuit.

Step 4 — Battery Sensing

For every port, the system measures: battery voltage, charging/discharging current, battery temperature, charging time, discharging time, charge capacity, discharge capacity, and number of cycles. These measurements are continuously supplied to the ESP32.

Step 5 — ESP32 Data Processing

The ESP32 reads the sensor data and determines the current operating condition of each port — for example, Battery 1: Voltage = 3.82 V, Current = 0.65 A, Temperature = 29°C, SOC ≈ 65%. The ESP32 also calculates accumulated charge/discharge using current over time.

Step 6 — Manual Mode

The user selects Manual Mode from the laptop/phone application. The dashboard shows live readings per port, for example:

Port	Voltage	Current	Temperature	SOC	Relay
1	3.82 V	0.65 A	29°C	65%	ON
2	3.55 V	0 A	27°C	35%	OFF
3	4.05 V	0.30 A	30°C	85%	ON

The user can press Port 1 → ON, Port 2 → OFF, Port 3 → ON, and the ESP32 operates the corresponding relays.

Step 7 — Automatic Mode

In Automatic Mode, the user sets thresholds — for example, start charging at 50% SOC and stop charging at 90% SOC. If SOC < 50%, the ESP32 activates the corresponding relay; when SOC ≥ 90%, it deactivates the relay. Temperature and safety limits can also be included.

Step 8 — Data Logging

The measured data are stored with timestamps (time, voltage, current, temperature, SOC, capacity, cycle), creating a battery history database.

Step 9 — Machine Learning Analysis

The collected battery data are supplied to an ML model, which determines battery condition, health, usable capacity, degradation trend, abnormal behaviour, charging-time prediction, and remaining performance — for example, Battery 1 → Healthy, Battery 2 → Normal, Battery 3 → Degraded. These classifications should be based on experimentally collected and properly labelled battery data rather than arbitrary thresholds.

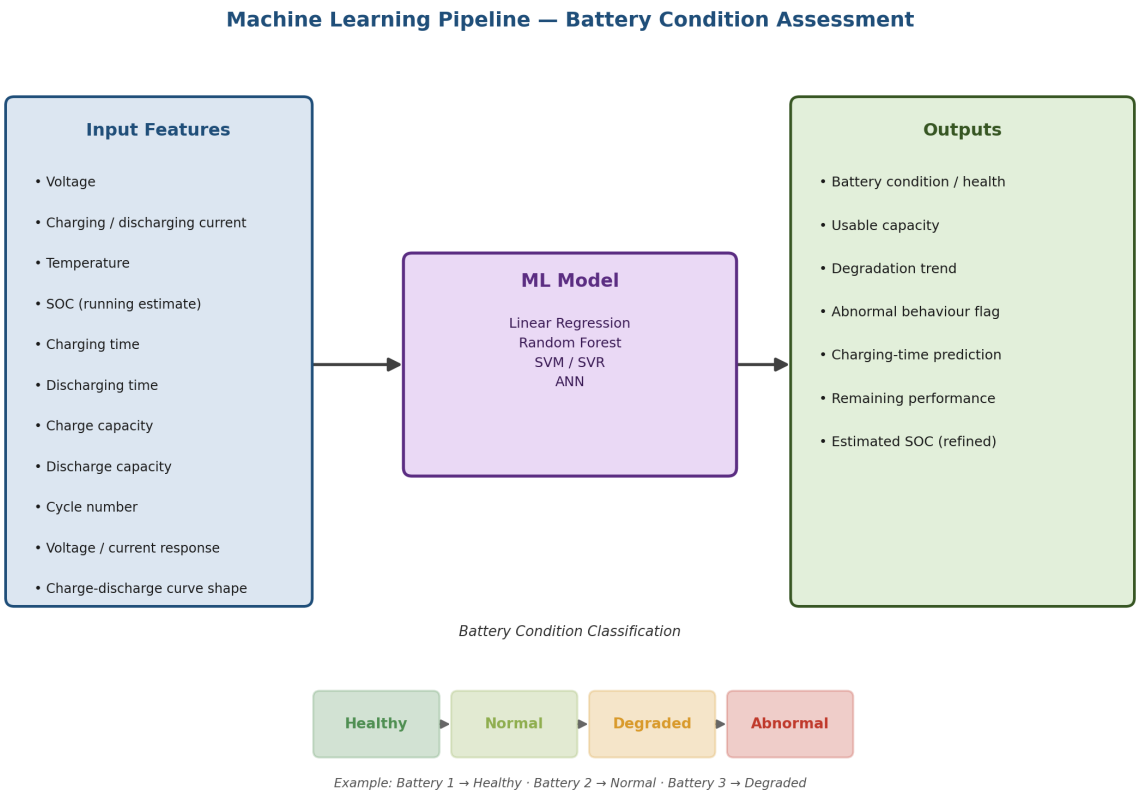


Figure 2: Machine learning pipeline — from measured battery parameters to condition classification.

4. Working Process — Per-Battery Cycle

The diagram below summarises the repeating cycle each battery goes through once connected: sensing, charge/discharge tracking, feature extraction, ML inference, dashboard update, and (where needed) a user alert. The cycle repeats continuously so that degradation trends are tracked across many cycles rather than judged from a single reading.

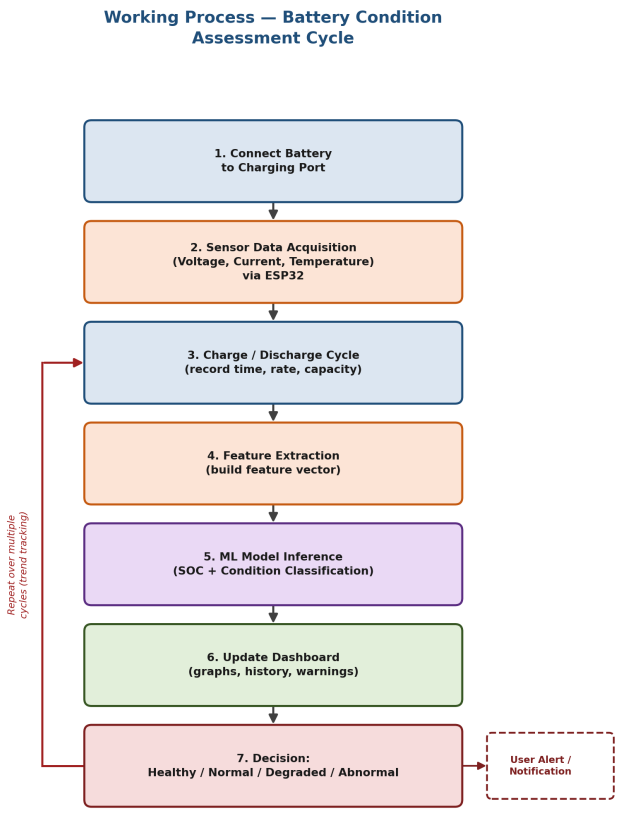


Figure 3: Working process of the battery condition assessment cycle.

5. Demo Software — User Dashboard

The mockup below illustrates the intended dashboard for the 3-port system. It shows live per-port voltage, current, temperature, SOC and relay status in Manual Mode; Automatic Mode threshold controls (start/stop SOC, temperature safety limit); the ML-based condition of each battery; and an SOC history chart across recent cycles for all three ports.

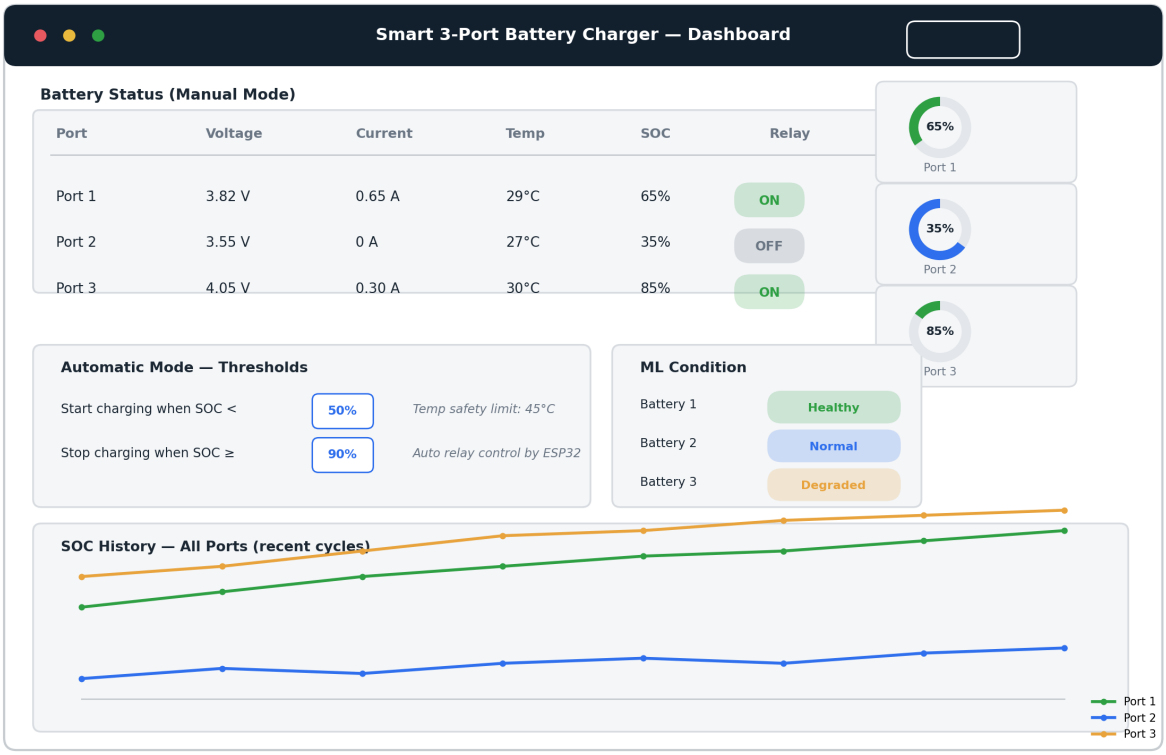


Figure 4: Mockup of the planned laptop/smartphone dashboard.

6. Equipment / Component List

For a 3-port prototype, a practical component list is:

Equipment	Quantity	Approx. Cost (৳)
ESP32 DevKit	1	500–700
AC-to-DC regulated power supply	1	700–1,200
Suitable CC/CV charging modules/stages	3	450–900
Relay modules / relay circuits	3	150–300
INA219/INA226 current sensor	3	360–750
Voltage sensing circuit	3	100–200
DS18B20 temperature sensor	3	150–300
Battery holders	3	120–250
Rechargeable batteries	3	450–900
Fuse + holders	3–4	100–200
Terminal blocks / connectors	—	150–250
PCB / perfboard	1	200–400
Wires / jumpers	—	150–250
OLED display (optional)	1	150–250

Enclosure	1	300–600
Miscellaneous components	—	200–400

Estimated prototype cost: approximately ৳3,800–৳6,700, depending mainly on the power supply, charging circuits, batteries, sensors, and enclosure. Prices are approximate Bangladesh-market estimates and can vary by seller and component quality.

7. Software Equipment

Software	Purpose
Arduino IDE	ESP32 programming
Python	Data processing + ML
Pandas	Battery dataset processing
NumPy	Numerical calculation
Matplotlib	Graphs
Scikit-learn	Machine Learning
SQLite / CSV	Data storage
Bluetooth / BLE	ESP32 ↔ phone/laptop
Flask / Streamlit	Dashboard

8. Main Features of the Final System

The completed prototype should provide:

1. 3 independent battery charging ports
2. Individual voltage measurement
3. Individual current measurement
4. Individual temperature measurement
5. Individual relay control
6. Manual charging mode
7. Automatic charging mode
8. User-defined start/stop thresholds
9. SOC estimation
10. Charge/discharge capacity calculation
11. Charging/discharging time measurement
12. Real-time laptop/phone monitoring
13. Historical graphs
14. Battery-cycle data recording
15. ML-based battery condition analysis

16. Battery degradation tracking
17. Abnormal battery detection
18. Charging-time prediction
19. Independent control of every port

9. Important Design Point

For this project, the intended charging architecture is:

AC → regulated DC supply → proper CC/CV charging stage → relay → battery

rather than:

AC → 3.7 V supply → relay → Li-ion battery

because the second arrangement is not a proper Li-ion charging system. If the intended battery is specifically a 3.7 V 18650 Li-ion cell, the next design step is to select the exact AC-to-DC supply and 3-port CC/CV charging circuit, then calculate the current rating and protection for each port.

10. Conclusion

By combining independent multi-port charging hardware with structured sensing and machine learning, the proposed system goes well beyond a conventional battery charger. It safely charges each battery using a proper CC/CV profile, gives the user full manual and automatic control over every port, and continuously assesses battery health, capacity, and degradation — turning routine charging into an intelligent, data-driven battery management platform.